

UDL grid for assessing the UDL conformity of the TOKADI educational robot

Instrument used in the teacher survey (evaluation)

Construction rationale

The grid was derived from the 35 considerations of the UDL Guidelines 3.0 (CAST 2024). In a first step, all considerations were reviewed against the criterion of direct technological addressability; nine considerations concerning purely curricular or pedagogical aspects were excluded. The remaining 26 items cover the three UDL principles: Engagement (9), Representation (7) and Action & Expression (10). Two researchers independently rated relevance and item wording; discrepancies were resolved through discussion. The grid was piloted with two teachers not included in the main sample.

Rating scale

Five-point Likert scale ranging from 1 = does not apply to 5 = fully applies. Each item is rated on its own statement; the checkpoint number refers to the corresponding consideration in the UDL Guidelines 3.0.

Reliability in the present sample (N = 10)

Overall grid $\alpha = .802$; Action & Expression $\alpha = .726$; Representation $\alpha = .544$; Engagement $\alpha = .317$. Principle scores are reported as means across the corresponding considerations and interpreted descriptively; they are not treated as psychometrically validated subscales.

Questionnaire

Checkpoint		Item	1 – does not apply	2	3	4	5 – fully applies
Engagement (9 items)							
7.1	a)	I perceive that TOKADI accommodates different preferences and abilities of students through varied input options (buttons, clapping, whistling).	☐	☐	☐	☐	☐
8.4	b)	TOKADI's feedback (LED lights, sounds, movement) appears motivating to me and supports the maintenance of students' attention.	☐	☐	☐	☐	☐
8.3	c)	I observe that TOKADI supports collaborative learning when several students work with it simultaneously.	☐	☐	☐	☐	☐
7.3	d)	The possibility of assembling and reconfiguring TOKADI fosters, in my view, playful learning and creative engagement.	☐	☐	☐	☐	☐
8.3	e)	I perceive that the shared use of TOKADI fosters a sense of community in the classroom.	☐	☐	☐	☐	☐
9.1	f)	I observe that TOKADI encourages students to formulate their own goals and approaches.	☐	☐	☐	☐	☐
9.2	g)	The anthropomorphic design (e.g. head, tail) facilitates, in my view, an emotional bond and empathic interaction.	☐	☐	☐	☐	☐
9.3	h)	I observe that TOKADI prompts students to reflect on their own actions.	☐	☐	☐	☐	☐
7.2	i)	I observe that TOKADI, through its feedback, encourages students to reflect on their own actions.	☐	☐	☐	☐	☐
Representation (7 items)							
1.2	a)	I observe that TOKADI presents information multimodally (e.g. visually, acoustically), addressing different modes of perception.	☐	☐	☐	☐	☐
2.1	b)	Symbols, lights and sounds are, in my view, intuitively designed and support the understanding of language and symbols.	☐	☐	☐	☐	☐

Checkpoint		Item	1 – does not apply	2	3	4	5 – fully applies
3.2	c)	TOKADI responds consistently to inputs, allowing students to grasp cause-and-effect relationships.	☐	☐	☐	☐	☐
1.1	d)	I appreciate the possibility of individually adjusting perceptual settings (e.g. volume, light intensity) in order to avoid over-stimulation.	☐	☐	☐	☐	☐
1.2	e)	The tactile components (e.g. buttons, 3D-printed parts) foster, in my observation, haptic perception.	☐	☐	☐	☐	☐
2.5	f)	The clear status feedback via the LED matrix (e.g. recording, success, error) helps students to understand the system state.	☐	☐	☐	☐	☐
2.3 / 2.4	g)	I perceive that TOKADI, through non-verbal elements (symbols, lights, sounds), can be used independently of language, thereby facilitating the participation of students with diverse linguistic backgrounds.	☐	☐	☐	☐	☐
Action & Expression (10 items)							
4.1	a)	I perceive that TOKADI allows different modes of operation (direct control, recording of sequences), thereby supporting varied learning pathways.	☐	☐	☐	☐	☐
5.1	b)	The language-independent control by means of symbols and tactile cues facilitates access for diverse groups of students.	☐	☐	☐	☐	☐
6.4	c)	TOKADI encourages students, in my observation, to experiment and to develop their own problem-solving strategies.	☐	☐	☐	☐	☐
6.1	d)	The recording function for movement sequences fosters strategic planning and problem-solving.	☐	☐	☐	☐	☐
6.5	e)	TOKADI's modular and adaptable technology makes it easier for me to reduce barriers in the classroom and to include different groups of students.	☐	☐	☐	☐	☐
4.2	f)	I see that TOKADI can be used in combination with additional materials (e.g. symbol, role and activity cards).	☐	☐	☐	☐	☐
4.1	g)	TOKADI's modular construction facilitates adaptation to different learning environments and needs.	☐	☐	☐	☐	☐
6.4	h)	TOKADI's progress and actions are clearly indicated, so that students can reflect on their own learning process.	☐	☐	☐	☐	☐
5.2	i)	I can adapt TOKADI's functions in order to support different learning objectives in the classroom.	☐	☐	☐	☐	☐
4.2	j)	The open-source documentation and building instructions facilitate low-threshold access and repairability.	☐	☐	☐	☐	☐